

Homework 6

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Problem 1.

(a) When $x > 0$, $f'(x) = \frac{3}{2}x^{\frac{1}{2}}$

And we can easily see $f'(0) = 0$

$$\begin{aligned}\|\nabla f(x) - \nabla f(0)\| &= \frac{3}{2}x^{\frac{1}{2}} \leq L\|x\| = Lx \\ \Rightarrow \frac{3}{2} &\leq Lx^{\frac{1}{2}}\end{aligned}$$

When $x \rightarrow 0^+$, there is no such finite $L > 0$ satisfies the formulation.

So f is not Lipschitz smooth.

(b)

$$\begin{aligned}x_1 &= x_0 - \bar{\alpha}f'(x_0) \text{ If } x_0 > 0, f'(x_0) = \frac{3}{2}x_0^{\frac{1}{2}} \\ \Rightarrow x_1 &= x_0 - \bar{\alpha}\frac{3}{2}x_0^{\frac{1}{2}} = x_0^{\frac{1}{2}}\left(x_0^{\frac{1}{2}} - \frac{3}{2}\bar{\alpha}\right) \\ f(x_1) &= x_0^{\frac{3}{4}}\left|x_0^{\frac{1}{2}} - \frac{3}{2}\bar{\alpha}\right|^{\frac{3}{2}} \\ f(x_1) - f(x_0) &= x_0^{\frac{2}{3}}\left(x_0^{\frac{3}{4}}\left|x_0^{\frac{1}{2}} - \frac{3}{2}\bar{\alpha}\right|^{\frac{3}{2}} - 1\right) \geq 0 \\ \Rightarrow x_0 &\leq \frac{9}{16}\bar{\alpha}\end{aligned}$$

Similarly, when $x_0 < 0$, we need $x_0 \leq \frac{9}{16}\bar{\alpha}$.

If $x_0 = 0$, since $f'(0) = 0$, $\forall x_1, f(x_1) \geq f(x_0)$.

Thus, for any constant step size $\bar{\alpha} > 0$, there exists an initial point x_0 such that $f(x_1) \geq f(x_0)$.

Problem 2.

(a) (i)

$$\nabla f(x) = 2Ax = \begin{bmatrix} 4x_1 \\ 10x_2 \end{bmatrix}$$

$$x_1 = x_0 - \alpha_0 \nabla f(x_0) = \begin{bmatrix} 0.6 \\ 0 \end{bmatrix}$$

(ii)

$$d_0 = -\nabla f(x_0) = \begin{bmatrix} -4 \\ -10 \end{bmatrix}$$

$$\alpha_0 = -\frac{d_0^T d_0}{2d_0^T A d_0} = \frac{116}{1064} = \frac{29}{266}$$

$$x_1 = x_0 + \alpha_0 \nabla f(x_0) = \begin{bmatrix} \frac{75}{133} \\ -\frac{12}{133} \end{bmatrix}$$

(iii) When $\alpha_0 = 1$,

$$z_1 = f(x_0 - \alpha_0 \nabla f(x_0)) = 423$$

$$z_2 = f(x_0) - \gamma \alpha_0 \|\nabla f(x_0)\|^2 = -51$$

Since $z_1 > z_2$, the Armijo condition is not satisfied.

When $\alpha_0 = \sigma = 0.2$,

$$z_1 = f(x_0 - \alpha_0 \nabla f(x_0)) = 5.08$$

$$z_2 = f(x_0) - \gamma \alpha_0 \|\nabla f(x_0)\|^2 = -4.6$$

Since $z_1 > z_2$, the Armijo condition is not satisfied.

When $\alpha_0 = \sigma^2 = 0.04$,

$$z_1 = f(x_0 - \alpha_0 \nabla f(x_0)) = 3.2112$$

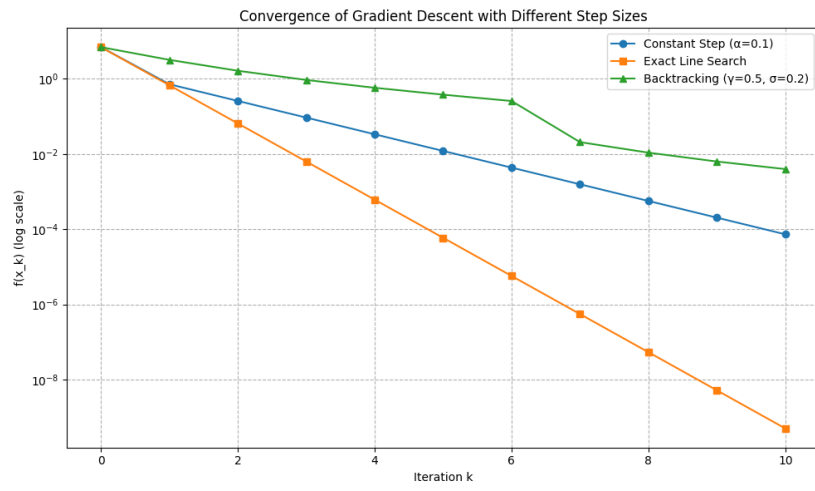
$$z_2 = f(x_0) - \gamma \alpha_0 \|\nabla f(x_0)\|^2 = 4.68$$

Since $z_1 < z_2$, the Armijo condition is satisfied.

Thus, the step size determined by the Armijo rule is $\alpha_0 = 0.04$,

and $x_1 = \begin{bmatrix} 0.84 \\ 0.6 \end{bmatrix}$.

(b) The code is in `p2.py`. The final outputs after running the code are as follows:



Problem 3.

(a)

$$\begin{cases} \frac{\partial f}{\partial x} = x^{\frac{1}{3}} + x + y = 0 \\ \frac{\partial f}{\partial y} = x + y = 0 \end{cases} \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases}$$

$$H = \begin{bmatrix} \frac{1}{3}x^{-\frac{2}{3}} + 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad H(0,0) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Since $H(0,0)$ is positive semidefinite and f is a convex function, $(0,0)$ is a global minimum.

(b)

$$d = -(\nabla^2 f(x, y))^{-1} \nabla f(x, y) = - \begin{bmatrix} 3x_k \\ -2x_k + y_k \end{bmatrix}$$

$$x_{k+1} = x_k + d_1 = -2x_k$$

$$y_{k+1} = y_k + d_2 = 2x_k$$

Thus, we can have $x_k \neq 0 \Rightarrow x_{k+1} \neq 0$, i.e., Newton's method will not converge if the initial point is not the optimal point.

(c) $\Omega = (x, y) : x + y = 1 = z : az = b$,

while $z = (x, y)^{-1}$, $a = (1, 1)^{-1}$, $b = 1$.

Take an unconstrained step:

$$z'_{k+1} = z_k - \alpha_k \nabla f(z_k) = - \begin{bmatrix} -\alpha_k x_k^{\frac{1}{3}} + (1 - \alpha_k)x_k - \alpha_k y_k \\ -\alpha_k x_k + (1 - \alpha_k)y_k \end{bmatrix}$$

$$\begin{aligned}
z_{k+1} &= P_{\Omega}(z'_{k+1}) \\
&= z'_{k+1} - \frac{a^T z'_{k+1} - b}{||a||^2} a \\
&= \frac{1}{2} \begin{bmatrix} -\alpha_k x_k^{\frac{1}{3}} + x_k - y_k - 1 \\ \alpha_k x_k^{\frac{1}{3}} - x_k + y_k - 1 \end{bmatrix}
\end{aligned}$$

$$\text{Thus, } \begin{bmatrix} x_{k+1} \\ y_{k+1} \end{bmatrix} = \begin{bmatrix} -\alpha_k x_k^{\frac{1}{3}} + x_k - y_k - 1 \\ \alpha_k x_k^{\frac{1}{3}} - x_k + y_k - 1 \end{bmatrix}.$$

Problem 4.

(a) Let $y = p + tv$, then

$$\begin{aligned}
 \min_y f &= \frac{1}{2} \|y - x\|^2 = \frac{1}{2} \|p + tv - x\|^2 \\
 &= \frac{\partial f}{\partial t} = v^T(p + tv - x) = 0 \\
 &\Rightarrow t^* = \frac{v^T(x - p)}{\|v\|^2} \\
 &\Rightarrow P_L(x) = y^* = p + \frac{v^T(x - p)}{\|v\|^2} v
 \end{aligned}$$

(b)

$$P_L(x) = \operatorname{argmin}_{y \in L} \|y - x\|^2$$

If $y^* = P_L(x) \in B$,

$$\begin{aligned}
 y^* &= \operatorname{argmin}_{y \in (L \cap B)} \|y - x\|^2 \\
 &= \operatorname{argmin}_{y \in C} \|y - x\|^2 \\
 &= P_C(x)
 \end{aligned}$$

Thus, when $P_L(x) \in B$, $P_C(x) = P_L(x)$.